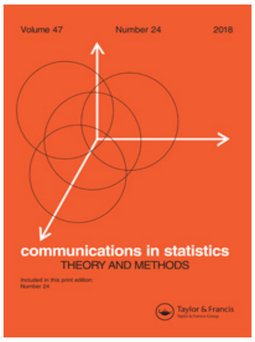

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On shifted integer-valued autoregressive model for count time series showing equidispersion, underdispersion or overdispersion

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ABSTRACT
In this paper, we introduce a first order integer-valued autoregressive process with Borel innovations based on the binomial thinning. This model is suitable to modeling zero truncated count time series with equidispersion, underdispersion and overdispersion. The basic properties of the process are obtained. To estimate the unknown parameters, the Yule-Walker (YW), conditional least squares (CLS) and conditional maximum likelihood (CML) methods are considered. The asymptotic distribution of CLS estimators are obtained and hypothesis tests to test an equidispersed model against an underdispersed or overdispersed model are formulated. A Monte Carlo simulation is presented analyzing the estimators performance in finite samples. Two applications to real data are presented to show that the Borel INAR(1) model is suited to model underdispersed and overdispersed data counts.

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Asymptotic normality; binomial thinning operator; borel distribution; integer-valued time series model

1. Introduction

In the last three decades, there has been a growing interest in discrete-valued time series models (Bourguignon and Vasconcellos 2015). A reason for this is the numerous situations where we find time series with small integer values. The number of patients in a hospital at a specific point of time and the number of persons in a queue waiting for service at a certain moment; the daily number of absent workers in a firm and the monthly number of ongoing strikes in a particular country, region or industry (Jung and Tremayne 2006); the number of different IP address (Weiß 2007) are examples of integer-valued data time series. The aim is to define a model for integer-valued time series that has covariance and properties similar structure to the continuous-valued autoregressive (AR) model.

McKenzie (1985) and Al-Osh and Alzaid (1987), independently, were the first to introduce an integer-valued autoregressive (INAR) process with Poisson marginals, Poisson INAR(1), based on the binomial thinning operator (Steutel and van Harn 1979). The equidispersion is the principal characteristic of the Poisson INAR(1) model, i.e., the stationary mean is equal to stationary variance, however many real-world data

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sets present overdispersion, variance larger than their mean, or underdispersion, variance smaller than their mean. For this reason, the Poisson INAR(1) model is not always suited for modeling integer-valued time series. In this sense, several models with overdispersion or underdispersion were defined in the recent years.

Ristić, Bakouch, and Nastić (2009) defined the negative binomial thinning operator and based on the new thinning operator introduced the overdispersed integer-valued time series model with geometric marginal. Weiß (2013) investigated some distributions with underdispersion and combined these models with the INAR(1) process to obtain a process that presents underdispersion. Schweer and Weiß (2014) considered the compound Poisson INAR(1) processes (CPINAR(1)), which is suitable for modeling data sets with overdispersion. For such CPINAR(1) processes, explicit results were derived for joint moments, the k-step-ahead distribution, the stationary distribution and mixing properties. Closed-form expressions for the asymptotic distribution of the dispersion index for CPINAR(1) processes was derived and developed a test to overdispersion based on the index of dispersion. Bourguignon and Weiß (2017) defined a new thinning operator that extends the binomial and negative binomial thinning operators. Based on the new thinning introduced the BerG-INAR(1) process, a stationary count data with Bernoulli-geometric marginals. The BerG distribution was obtained as a convolution of the Bernoulli and geometric distributions. the BerG-INAR(1) process is suitable for time series data with equidispersion, underdispersion and overdispersion. They derived the probability generating function, moments, transition probabilities and zero probability. The maximum likelihood method was used for estimating the model parameters.

Although several first order non-negative integer valued time series models have been developed in recent years, there are some cases where are more suitable truncated models, for instance, integer-valued time series where zeros are not observed. The truncated models have not received much attention in research of count data times series, few works have been done in this case. Bakouch and Ristić (2010) introduced a new stationary INAR(1) process with zero truncated Poisson marginal distribution [ZTPINAR(1)]. Several properties of the process were derived and applications to two crime data sets were presented. Nastić (2012) defined two types of shifted geometric INAR(1) models based on the negative binomial thinning operator. These models are suitable to positive count data and have a simple form with only two parameters. Their correlation properties are derived, conditional mean and conditional variance are considered. Statistical properties of the innovations process are obtained, non-parametric estimators of models parameters are defined and their asymptotic characterizations are givens. Bourguignon and Vasconcellos (2015) introduced a family of INAR(1) processes with power series innovations (PSINAR) which is suitable to modeling non-negative integer valued time series with equidispersion, overdispersion and underdispersion. The PSINAR(1) models has the flexibility of modeling positive integers data sets, this can be achieved by considering truncated distributions for the process innovations.

The purpose of this paper is to introduce a shifted INAR(1) model with Borel innovations based on binomial thinning operator which exhibits equidispersion, overdispersion and underdispersion. The Borel INAR(1) (BINAR(1)) is a simple model with only two parameters suitable to zero truncated data sets. The process is an ergodic stationary Markov chain and its structure is different from other already proposed models.

Although the BINAR(1) process in some cases shows equidispersion, in this cases, it does not coincide with the Poisson INAR(1) model.

The paper is organized as follows. In Section 2, we introduce the Borel distribution proposed by Borel (1942), present its definition, properties and moments. Section 3 introduces the stationary Borel INAR model, its statistical properties are obtained, the transition probabilities, conditional mean and variance are derived. In Section 4, the parameter estimation is discussed; we consider CLS, YW and conditional maximum likelihood estimators. A simulation study is considered in Section 5. Two real data examples are presented in Section 6. Finally, we conclude the paper in Section 7.

2. Borel distribution

In this section, we will make a brief review about the Borel distribution (Borel 1942). We present its definition and properties. Later in Section 3, we will consider the Borel distribution to introduce the Borel INAR(1) model with Borel distribution innovations which is adequate for truncated data set with equidispersion, overdispersion or underdispersion.

2.1. Definition and properties

The Borel distribution is a discrete probability distribution that was first obtained by Borel (1942). It is included in a subclass of a family of discrete probability distributions known as Lagrangian probability distributions and is classified within the so called basic Lagrangian distributions of the first kind. The Borel distribution is a special case of the queuing process, it can be viewed as a particular case of the Borel-Tanner distribution (Haight and Breuer 1960). For more details of Lagrangian distributions see Consul and Famoye (2006).

The probability mass function (pmf) from a discrete random variable Y with Borel distribution is given by

$$\mathrm{P}(Y=y)=\frac{(y\lambda)^{y-1}\mathrm{e}^{-\lambda y}}{y!},\quad y=1,2,3,\ldots \tag{1}$$

where $0<\lambda<1$.

The Borel distribution has a simple interpretation in the theory of queues following Haight and Brauer (1960): "Suppose a queue is initiated with one member and let λ be equal to the traffic intensity assuming Poisson arrivals and constant service time. The pmf of Borel distribution, $\mathrm{P}(Y=y)$, represents the probability that exactly y members of queue will be served before the queue first vanishes".

The properties of the Borel distribution can be found in Section 8.4 from Consul and Famoye (2006). In particular, the mean and variance of the distribution in (1) are given by

$$\mu_Y=(1-\lambda)^{-1}\quad\text{and}\quad\sigma_Y^2=\lambda(1-\lambda)^{-3}$$

The Borel distribution satisfies the properties of equidispersion, when $\lambda=3/2-\sqrt{5}/2$, underdispersion, when $0<\lambda<3/2-\sqrt{5}/2$ and overdispersion, when $3/2-\sqrt{5}/2<\lambda<1$.

We have that the probability generating function (pgf), $E(u^Y) := G_Y(u)$, and moment generating function (mgf), $M_{Y,}$ of a random variable Y following the Borel distribution are, respectively, given by

$$G_Y(u) = z, \quad \text{where} \quad z = u \ \mathrm{e}^{\lambda(z-1)}$$

and

$$M_Y(\beta) = \mathrm{e}^s, \quad \text{where} \quad s = \beta + \lambda(\mathrm{e}^s - 1)$$

All the moments of the Borel distribution exist for $0 < \lambda < 1$. The kth noncentral moment, $\mu'_k = E(Y^k)$, and the central moment, $\mu_k = E(Y - \mu_Y)^k$, satisfy the following recurrence relations,

$$\mu'_{k+1} = \lambda(1-\lambda)^{-1}\frac{d\mu'_k}{d\lambda} + \mu'_1\mu'_k, \quad k = 0, 1, 2, \ldots$$

and

$$\mu_{k+1} = \lambda(1-\lambda)^{-1}\frac{d\mu_k}{d\lambda} + k\mu_2\mu_{k-1}, \quad k = 1, 2, 3, \ldots$$

where $\mu'_1 = (1-\lambda)^{-1}$ and $\mu_2 = \sigma^2_Y$.

In particular, the third and fourth central moments are given by

$$\mu_3 = \lambda(1+2\lambda)(1-\lambda)^{-5} \quad \text{and} \quad \mu_4 = \lambda(1+8\lambda+6\lambda^2)(1-\lambda)^{-7} + 3\mu_2^2 \tag{2}$$

Observe that, in the particular case that one the Borel distribution is equidispersed, where $\lambda = \lambda_0 = 3/2 - \sqrt{5}/2$, we have $\mu_Y = \sigma^2_Y = 1/(1-\lambda_0)$ but different from μ_3 given in (2). Furthermore, the Poisson distribution with parameter $1/(1-\lambda_0)$ have mean, variance and μ_3 equal to $1/(1-\lambda_0)$ being different of the equidispersed Borel distribution.

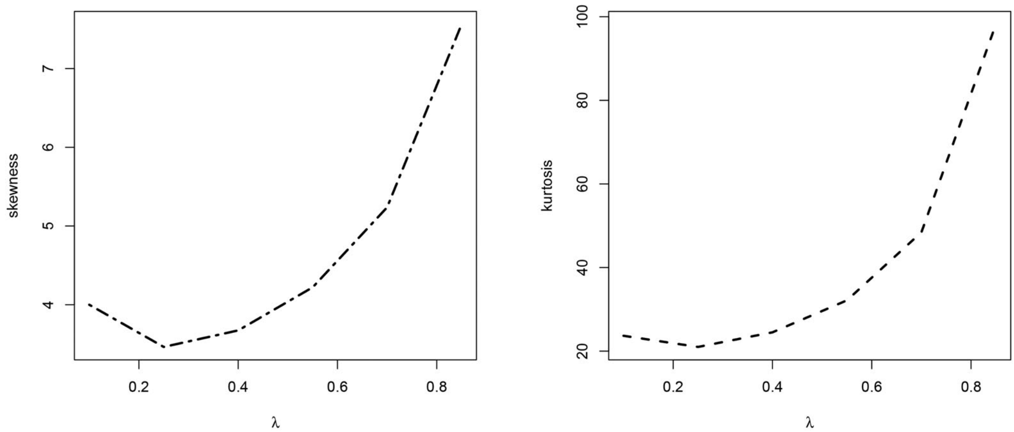
The coefficients of skewness, β_1, and kurtosis, β_2, are given by the expressions

$$\beta_1 = \frac{1+2\lambda}{\sqrt{\lambda(1-\lambda)}} \quad \text{and} \quad \beta_2 = 3 + \frac{1+8\lambda+6\lambda^2}{\lambda(1-\lambda)} \tag{3}$$

Figure 1 displays the skewness and kurtosis of Borel distribution. From (3) and Figure (1), we have that the Borel distribution is always positively skewed and leptokurtic.

3. Borel INAR(1) model

Let $\mathbb{Z}$ be the set of all integers, $\mathbb{N}$ be the set of positive integers and $\mathbb{N}_0$ the set of non-negative integers. In this section, we will introduce the first order integer-valued autoregressive process with Borel distribution innovations on $\mathbb{N}$, Borel INAR(1). This model is suitable for positive counting processes with equidispersion, overdispersion and underdispersion. Basic properties of the process are derived. The autocorrelation function, transition probabilities, conditional mean and variance are obtained.



(a) (b)

Figure 1. Skewness and kurtosis of Borel distribution. (a) Skewness of Borel distribution (b) Kurtosis of Borel distribution.

3.1. Definition and properties of the process

Let X be a non-negative integer valued random variable and $\alpha \in [0, 1]$. Steutel and van Harn (1979) introduced the binomial thinning operator as

$$\alpha \circ X = \sum_{i=1}^{X} W_i \tag{4}$$

where W_i are independent and identically distributed binary random variables, independent of X with $P(W_i = 1) = 1 - P(W_i = 0) = \alpha$, i.e., W_i has Bernoulli distribution. the sequence $\{W_i\}_{i\in\mathbb{N}}$ is called the counting series for $\alpha \circ X$. The properties of the binomial thinning operator can be found in Silva and Oliveira (2004) and Silva (2005). Based on this operator, we define the first-order integer-valued autoregressive process with Borel innovations.

Definition 3.1. *(Borel INAR(1)) The first-order integer-valued autoregressive process with Borel innovations $\{X_t\}_{t\in\mathbb{Z}}$, BINAR(1), is defined by equation*

$$X_t = \alpha \circ X_{t-1} + \epsilon_t, \quad t \in \mathbb{Z} \tag{5}$$

where $\alpha \in [0, 1), \alpha \circ X_{t-1}$ is the binomial thinning operator in (4), $\{\epsilon_t\}_{t\in\mathbb{Z}}$ is the sequence of innovations with Borel(λ) distribution and ϵ_t is independent of W_{t-i} for all $i \geq 1$. Since $\alpha \in [0, 1)$ and $\mu_\epsilon, \sigma_\epsilon^2 < \infty$, from Du and Li (1991), the $\{X_t\}_{t\in\mathbb{Z}}$ process in (5) is an ergodic stationary Markov chain with transition probabilities given by

$$\Pr(X_t = k | X_{t-1} = l) = \sum_{i=0}^{\min(k-1,l)} \binom{l}{i} \alpha^i (1-\alpha)^{l-i} \frac{[(k-i)\lambda]^{k-i-1}}{(k-i)!} \mathrm{e}^{-\lambda(k-i)} \tag{6}$$

It is easy to see that, if $\mu_\epsilon, \sigma_\epsilon^2 < \infty$ the mean and variance of the INAR(1) process are given by

$$\mu_X = \mu_\epsilon/(1-\alpha) \quad \text{and} \quad \sigma_X^2 = (\sigma_\epsilon^2 + \alpha\mu_\epsilon)/(1-\alpha^2)$$

where $\mu_\epsilon, \sigma_\epsilon^2$ are the mean and variance of innovations respectively. Thus, in the BINAR(1) process, we have

$$\mu_X = \mathrm{E}(X_t) = \frac{1}{(1-\alpha)(1-\lambda)} \qquad \text{and} \qquad \sigma_X^2 = \mathrm{Var}(X_t) = \frac{\alpha(1-\lambda)^2 + \lambda}{(1-\alpha^2)(1-\lambda)^3}$$

The dispersion index of X_t in INAR(1) model, I_X, is directly linked with the dispersion index of the innovations (Schweer and Weiß 2014), $I_\epsilon = \sigma_\epsilon^2/\mu_\epsilon$, by expression

$$I_X = \frac{\sigma_X^2}{\mu_X} = \left(1 + \frac{I_\epsilon}{\alpha}\right) \cdot \left(1 + \frac{1}{\alpha}\right)^{-1}$$

Thus, we have that the dispersion of the process depends of the innovations dispersion. The BINAR(1) dispersion index (based on the stationary mean and variance) is given by

$$I_X = \frac{\lambda/(1-\lambda)^2 + \alpha}{1+\alpha};$$

since we have Borel innovations, it follows that this model presents

- equidispersion when $\lambda = 3/2 - \sqrt{5}/2$,
- overdispersion when $3/2 - \sqrt{5}/2 < \lambda < 1$,
- underdispersion when $0 < \lambda < 3/2 - \sqrt{5}/2$.

We have that the Borel INAR(1) model is suited when the time series data presents overdispersion or underdispersion.

The conditional mean and variance of X_t given X_{t-1} is

$$\mathrm{E}(X_t|X_{t-1}) = \alpha X_{t-1} + \frac{1}{(1-\lambda)}$$

and

$$\mathrm{Var}(X_t|X_{t-1}) = \alpha(1-\alpha)X_{t-1} + \frac{3\ \lambda - 1 - \lambda^2}{(1-\lambda)^3} + \frac{1}{(1-\lambda)}$$

It is also easy to verify that the autocorrelation function (ACF) at lag k is given by

$$\mathrm{Corr}(X_t, X_{t-k}) = \rho(k) = \alpha^k, \;\; k \geq 1 \tag{7}$$

which clearly is restricted to be positive and have the same behavior as that of the AR(1) model.

4. Estimation and inference of the unknown parameters

In practice, the true values of the model parameters α and λ are not known but have to be estimated from a given time series data. This section is concerned with the estimation of the two parameters of interest. We consider three estimation methods, namely, conditional least squares (CLS), Yule-Walker (YW) and conditional maximum likelihood (CML).

4.1. Conditional least squares estimation

The CLS estimator $\hat{\boldsymbol{\theta}} = (\hat{\alpha}, \hat{\lambda})^{\mathsf{T}}$ of $\boldsymbol{\theta} = (\alpha, \lambda)^{\mathsf{T}}$ is given by

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} (Q_T(\boldsymbol{\theta}))$$

where $Q_T(\boldsymbol{\theta}) = \sum_{t=2}^{T} [X_t - E(X_t|X_{t-1})]^2$. Thus, following Klimko and Nelson (1978), the CLS estimators of α and λ can be written in closed form as

$$\hat{\alpha}_{\mathrm{CLS}} = \frac{\sum_{t=2}^{T} X_t X_{t-1} - \frac{1}{T-1} \sum_{t=2}^{T} X_t \sum_{t=2}^{T} X_{t-1}}{\sum_{t=2}^{T} X_{t-1}^2 - \frac{1}{T-1} \left(\sum_{t=2}^{T} X_{t-1}\right)^2} \tag{8}$$

and

$$\hat{\lambda}_{\mathrm{CLS}} = 1 - \frac{T-1}{\left(\sum_{t=2}^{T} X_t - \hat{\alpha}_{\mathrm{CLS}} \sum_{t=2}^{T} X_{t-1}\right)}$$

where $\hat{\alpha}_{\mathrm{CLS}}$ is given in (8).

We establish the asymptotic normality of CLS estimators in the following proposition which proof can be found in the Appendix.

Theorem 4.1. *The CLS estimators $\hat{\alpha}_{\mathrm{CLS}}$ and $\hat{\lambda}_{\mathrm{CLS}}$ are strongly consistent and asymptotic normal, i.e.,*

$$T^{1/2}(\hat{\alpha}_{\mathrm{CLS}} - \alpha, \hat{\lambda}_{\mathrm{CLS}} - \lambda) \xrightarrow{d} N(0, \boldsymbol{\Sigma})$$

with

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_\alpha^2 & \sigma_{\alpha\lambda}^2 \\ \sigma_{\alpha\lambda}^2 & \sigma_\lambda^2 \end{pmatrix}$$

where

$$\sigma_\alpha^2 = (1-\alpha^2) + \frac{\alpha(1-\alpha)(1-\alpha^2)^2(1-\lambda)^5}{(1-\alpha^3)(\lambda+\alpha(1-\lambda)^2)^2} \left[\frac{\alpha}{1-\alpha} + \frac{3\alpha^3}{1+\alpha} + \frac{3\alpha^2\lambda}{(1+\alpha)(1-\lambda)^2} + \frac{1+2\lambda}{(1-\lambda)^4} \right]$$

$$\begin{aligned}
\sigma_{\alpha\lambda}^2 = {} & \alpha(1-\alpha)(1-\lambda)^2 - \frac{\lambda(1-\alpha)(1-\lambda)}{\lambda+\alpha(1-\lambda)^2} - \frac{\alpha(1+\alpha)(1-\lambda)^3}{\lambda+\alpha(1-\lambda)^2} \\
& - \left[(1+\alpha) + 3\alpha^2(1-\alpha)\right] \frac{\alpha^2(1-\alpha^2)(1-\lambda)^7}{\left[\lambda+\alpha(1-\lambda)^2\right]^2} \\
& - \left[3\alpha^2\lambda(1-\lambda)^2 + (1+2\lambda)(1+\alpha)\right] \frac{\alpha(1-\alpha)^2(1+\alpha)(1-\lambda)^3}{\left[\lambda+\alpha(1-\lambda)^2\right]^2}
\end{aligned}$$

$$\begin{aligned}
\sigma_\lambda^2 = {} & \left\{1 - \alpha\left[\alpha\lambda + (1-2\alpha)(1+\lambda^2) + \alpha^2(1-\lambda)^2\right]\right\} \frac{1-\lambda}{(1-\alpha)^2} \\
& + \frac{\alpha(1+\alpha)(1-\lambda)^4}{(1-\alpha)\left[\lambda+\alpha(1-\lambda)^2\right]} + \frac{3\alpha^3(1-\lambda)^5(1-\alpha^2)}{(1-\alpha^3)\left[\lambda+\alpha(1-\lambda)^2\right]} \\
& + \left[(1-\alpha)(1+2\lambda) + \alpha(1-\lambda)^4\right] \frac{\alpha(1+\alpha)^2(1-\lambda)^3}{(1-\alpha^3)\left[\lambda+\alpha(1-\lambda)^2\right]^2}
\end{aligned}$$

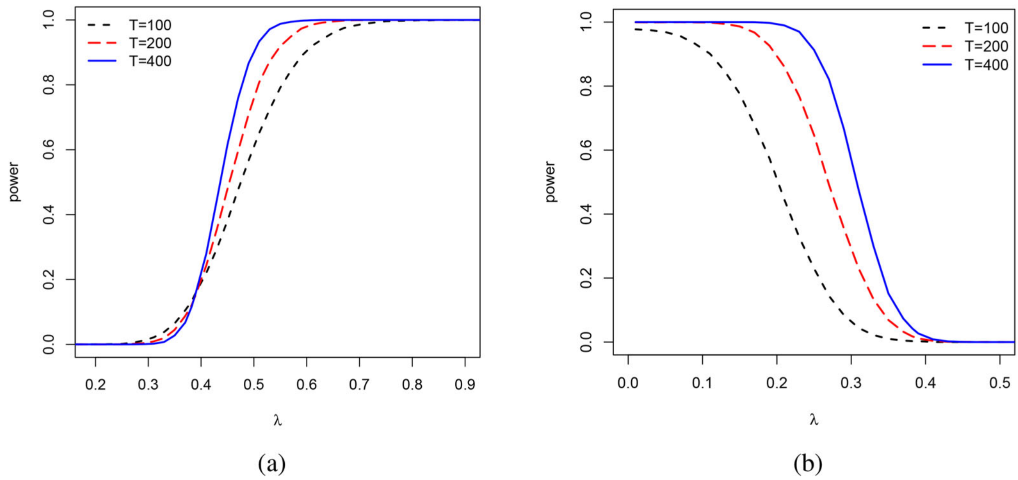


Figure 2. Asymptotic power against λ for BINAR(1) model. (a) H_1: $\lambda > \lambda_0$ (b) H_1: $\lambda < \lambda_0$.

Considering $\lambda_0 = 3/2 - \sqrt{5}/2$, the asymptotic normality of $\hat{\lambda}_{\text{CLS}}$ is particularly useful if our interest is to test the null hypotheses of an equidispersed model, $H_0 : \lambda = \lambda_0$, against the alternative hypotheses of an underdispersed model $H_1 : \lambda < \lambda_0$ or an overdispersed model, $H_1 : \lambda > \lambda_0$.

Based on the asymptotic normality of $\hat{\lambda}_{CLS}$ estimator, we will reject $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda < \lambda_0$ on significance level δ if

$$\hat{\lambda}_{CLS} < \lambda_0 + q_\delta \cdot \sqrt{\hat{\sigma}^2_{\lambda_0}/T} \tag{9}$$

and we will reject $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda > \lambda_0$ on significance level δ if

$$\hat{\lambda}_{CLS} > \lambda_0 + q_{1-\delta} \cdot \sqrt{\hat{\sigma}^2_{\lambda_0}/T} \tag{10}$$

where $\hat{\sigma}^2_{\lambda_0}$ is the variance of λ_0 obtained replacing λ_0 in the σ^2_λ expression, q_δ is the δ-quantile of the standard normal distribution, this is, $\Phi(q_\delta) = \delta, \delta \in (0,1)$ and Φ is the standard normal distribution. Alternatively, we will reject $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda < \lambda_0$ (overdispersion) if

$$\Phi\left((\hat{\lambda} - \lambda_0)/\sqrt{\hat{\sigma}^2_{\lambda_0}/T}\right) < \delta$$

If the alternative hypothesis of interest is $H_1 : \lambda > \lambda_0$ (underdispersion), we will reject $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda > \lambda_0$ when

$$1 - \Phi\left((\hat{\lambda} - \lambda_0)/\sqrt{\hat{\sigma}^2_{\lambda_0}/T}\right) < \delta$$

Figure 2 shows the asymptotic behavior of the power functions against λ. We consider the significance level $\delta = 0.05, \alpha = 0.15$ and $\lambda \in (0,1)$. We can see, as expected, that the power of dispersion test becomes better as we increase the sample size T.

Table 1. Simulated Borel INAR(1) with $(\alpha, \lambda) = (0.2, 0.2)$, significance level $\delta = 0.05$ and 10.000 replicas. Empirical and asymptotic properties of $\hat{\lambda}$ and empirical rejection rates (r.r.).

T	Mean	s.d.	s.d.$_a$	Skewness	$\hat{q}_{0.05}$	$q_{0.05,a}$	$\hat{q}_{0.95}$	$q_{0.95,a}$	r.r.$_\alpha$	r.r.$_{\hat{\alpha}}$	r.r.$_a$
100	0.2012	0.1168	0.1066	−0.9253	−0.0100	0.0245	0.3639	0.3754	0.4688	0.4764	0.4660
250	0.2003	0.0731	0.0674	−0.5094	0.0672	0.0890	0.3085	0.3109	0.8597	0.8894	0.8375
500	0.2021	0.0524	0.0476	−0.3405	0.1114	0.1215	0.2833	0.2784	0.9874	0.9927	0.9837

Tables 1–3 present results from a simulation to access the quality of asymptotic approximations in Theorem 4.1. We simulated the underdispersed Borel INAR(1) with $(\alpha, \lambda) = (0.2, 0.2)$, overdispersed Borel INAR(1) with $(\alpha, \lambda) = (0.2, 0.5)$ and equidispersed Borel INAR(1) with $(\alpha, \lambda) = \left(0.2, 3/2 - \sqrt{5}/2\right)$, respectively. We considered the significance level $\delta = 0.05$ and 10.000 replicas in both cases. In the same way of Schweer and Weiß (2014), we show both the empirically observed results and the values obtained from the asymptotic approximation.

The rates of rejecting the null hypothesis of a Borel INAR(1) with parameters (α, λ) are presented in both tables, where the critical value according to (9) in the underdispersed case and according to (10) in the overdispersed case was computed either with the true α (r.r.$_\alpha$) or by plugging-in CLS estimator $\hat{\alpha}$ (r.r.$_{\hat{\alpha}}$). We observe, in both cases, that the reject rate increases as we increase the sample size, to both for α and $\hat{\alpha}$ and the asymptotic approximating (r.r.$_a$) and the empirical values of reject rates are very close to the ones in the asymptotic approximate. In general, the empirical results of the quantiles $\hat{q}_\beta$ with $\beta = 0.05, 0.95$ and the standard deviation is very close to its asymptotic results for $T \geq 250$ as expected. We conclude that the asymptotic results have reasonable approximations for finite samples in both, underdispersed and overdispersed cases.

4.2. Yule-Walker estimation

Let $X_1, \ldots, X_T$ stem from a stationary BINAR(1) process $\{X_t\}_{t\in\mathbb{Z}}$, the sample autocorrelation function is given by

$$\hat{\rho}(k) = \frac{\sum_{t=1}^{T-k}(X_t - \bar{X})(X_{t+k} - \bar{Y})}{\sum_{t=1}^{n}(X_t - \bar{X})^2}$$

where $\bar{X} = (1/T)\sum_{t=1}^{T} X_t$ is the sample mean. The Yule-Walker (YW) estimator of α, based upon the fact that $\rho(k) = \alpha^k$, as in (7), is given by

$$\hat{\alpha}_{\mathrm{YW}} = \hat{\rho}(1) = \frac{\sum_{t=1}^{T-1}(X_t - \bar{X})(X_{t+1} - \bar{X})}{\sum_{t=1}^{T}(X_t - \bar{X})^2} \tag{11}$$

The first moment of $\{X_t\}_{t\in\mathbb{Z}}$ is given by $\mathrm{E}(X_t) = 1/(1-\alpha)(1-\lambda)$. Using this, the estimator of λ is defined as

$$\hat{\lambda}_{\mathrm{YW}} = 1 - \frac{1}{\bar{X}(1 - \hat{\alpha}_{\mathrm{YW}})}$$

where $\hat{\alpha}_{\mathrm{YW}}$ is given in (11).

Table 2. Simulated Borel INAR(1) with $(\alpha, \lambda) = (0.2, 0.5)$, significance level $\delta = 0.05$ and 10.000 replicas. Empirical and asymptotic properties of $\hat{\lambda}$ and empirical rejection rates.

T	Mean	s.d.	s.d.$_a$	Skewness	$\hat{q}_{0.05}$	$q_{0.05,a}$	$\hat{q}_{0.95}$	$q_{0.95,a}$	r.r.$_\alpha$	r.r.$_{\hat{\alpha}}$	r.r.$_a$
100	0.4970	0.0840	0.0821	−1.0474	0.3421	0.3648	0.6351	0.6100	0.4647	0.4781	0.4527
250	0.4980	0.0522	0.0519	−0.6039	0.4052	0.4145	0.5753	0.5854	0.7303	0.7189	0.7469
500	0.4998	0.0376	0.0367	−0.4280	0.4343	0.4395	0.5569	0.5604	0.9150	0.9003	0.9272

Table 3. Simulated Borel INAR(1) with $(\alpha, \lambda) = \left(0.2, 3/2 - \sqrt{5}/2\right)$, significance level $\delta = 0.05$ and 10.000 replicas. Empirical and asymptotic properties of $\hat{\lambda}$ and empirical rejection rates.

T	Mean	s.d.	s.d.$_a$	Skewness	$\hat{q}_{0.05}$	$q_{0.05,a}$	$\hat{q}_{0.95}$	$q_{0.95,a}$	r.r.$_\alpha$	r.r.$_{\hat{\alpha}}$	r.r.$_a$
100	0.3814	0.0963	0.0925	−0.9560	0.2055	0.2297	0.5162	0.5341	0.0528	0.0754	0.0244
250	0.3821	0.0604	0.0585	−0.6021	0.2747	0.2857	0.4703	0.4782	0.0504	0.0660	0.0342
500	0.3821	0.0431	0.0413	−0.4306	0.3067	0.3139	0.4469	0.4500	0.0539	0.0659	0.0418

Freeland and McCabe (2005) considering a Poisson INAR(1) process showed that the YW and CLS estimators are asymptotic equivalent. The next proposition, which proof is in the Appendix, shows that the result holds to the BINAR(1) process.

Theorem 4.2. *In the BINAR(1) process*

$$\begin{pmatrix} \hat{\alpha}_{\text{CLS}} - \hat{\alpha}_{\text{YW}} \\ \hat{\lambda}_{\text{CLS}} - \hat{\lambda}_{\text{YW}} \end{pmatrix} = o_p(T^{-1/2})$$

and this is sufficient for the CLS and YW estimators to have the same asymptotic distribution.

In Section 5 we compare CLS and YW estimators, we can see that the results of both are similar as expected.

4.3. Conditional maximum likelihood estimation

Suppose that X_1 is fixed. The conditional log-likelihood function is then given by

$$\ell(\boldsymbol{\theta}) = \log\left[\prod_{t=2}^{T} \Pr(X_t = k | X_{t-1} = l)\right] = \sum_{t=2}^{T} \log\left[\Pr(X_t = k | X_{t-1} = l)\right]$$

with $\Pr(X_t = k | X_{t-1} = l)$ as in (6).

The CML estimator of $\boldsymbol{\theta}$ is the value $\hat{\boldsymbol{\theta}}_{\text{CML}} = (\hat{\alpha}_{\text{CML}}, \hat{\lambda}_{\text{CML}})^{\top}$ that maximizes $\ell(\boldsymbol{\theta})$. Since $\ell'(\boldsymbol{\theta})$ is a nonlinear function, the maximum likelihood estimate of $\boldsymbol{\theta}$ must be computed by using numerical methods.

CML estimators, as commented above, do not have closed-form expressions and Fisher's information matrix, $\mathbf{K}(\boldsymbol{\theta})$, is also not available. In practice, the observed Fisher information $(T-1)\mathbf{J}(\boldsymbol{\theta})$ can be used to approximate $\mathbf{K}(\boldsymbol{\theta})$, and plugging in the obtained estimates allows, for instance, to approximate the asymptotic standard errors of the CML estimators; also see Freeland and McCabe (2004).

5. Estimators numerical comparison

In this section, it was considered a Monte Carlo simulation experiment to study the behavior of the CLS, YW and CML estimators. The Monte Carlo simulation

Table 4. Empirical means and mean squared errors (in parentheses) of the estimates of the parameters for $\lambda = 0.3$ and some values of α and T.

		Estimator of α			Estimator of λ		
T	α	$\hat{\alpha}_{YW}$	$\hat{\alpha}_{CLS}$	$\hat{\alpha}_{CML}$	$\hat{\lambda}_{YW}$	$\hat{\lambda}_{CLS}$	$\hat{\lambda}_{CML}$
50	0.15	0.1171	0.1196	0.1554	0.3002	0.2968	0.2833
		(0.0210)	(0.0217)	(0.0050)	(0.0186)	(0.0197)	(0.0071)
	0.30	0.2530	0.2588	0.3024	0.3101	0.3029	0.2832
		(0.0226)	(0.0230)	(0.0046)	(0.0234)	(0.0259)	(0.0076)
	0.45	0.3884	0.3968	0.4489	0.3303	0.3186	0.2859
		(0.0218)	(0.0214)	(0.0037)	(0.0280)	(0.0301)	(0.0073)
	0.60	0.5277	0.5396	0.5996	0.3556	0.3345	0.2859
		(0.0212)	(0.0201)	(0.0023)	(0.0387)	(0.0431)	(0.0073)
	0.75	0.6633	0.6797	0.7485	0.4150	0.3743	0.2895
		(0.0199)	(0.0175)	(0.0010)	(0.0565)	(0.0677)	(0.0070)
100	0.15	0.1319	0.1332	0.1532	0.3017	0.3005	0.2931
		(0.0115)	(0.0116)	(0.0025)	(0.0102)	(0.0104)	(0.0034)
	0.30	0.2746	0.2776	0.3016	0.3069	0.3037	0.2923
		(0.0116)	(0.0116)	(0.0023)	(0.0121)	(0.0126)	(0.0035)
	0.45	0.4192	0.4237	0.4506	0.3147	0.3088	0.2935
		(0.0107)	(0.0106)	(0.0017)	(0.0164)	(0.0171)	(0.0034)
	0.60	0.5643	0.5702	0.5994	0.3261	0.3156	0.2933
		(0.0091)	(0.0087)	(0.0011)	(0.0223)	(0.0235)	(0.0034)
	0.75	0.7066	0.7143	0.7486	0.3585	0.3383	0.2948
		(0.0077)	(0.0071)	(0.0005)	(0.0333)	(0.0355)	(0.0035)
200	0.15	0.1415	0.1421	0.1513	0.2995	0.2989	0.2960
		(0.0060)	(0.0060)	(0.0012)	(0.0050)	(0.0051)	(0.0017)
	0.30	0.2872	0.2887	0.3000	0.3030	0.3015	0.2972
		(0.0060)	(0.0061)	(0.0011)	(0.0066)	(0.0068)	(0.0017)
	0.45	0.4349	0.4371	0.4497	0.3064	0.3034	0.2968
		(0.0052)	(0.0052)	(0.0009)	(0.0081)	(0.0083)	(0.0018)
	0.60	0.5825	0.5855	0.5995	0.3121	0.3070	0.2972
		(0.0043)	(0.0042)	(0.0006)	(0.0117)	(0.0120)	(0.0017)
	0.75	0.7289	0.7328	0.7495	0.3281	0.3174	0.2962
		(0.0031)	(0.0029)	(0.0003)	(0.0180)	(0.0188)	(0.0018)
400	0.15	0.1456	0.1460	0.1506	0.3001	0.2998	0.2984
		(0.0030)	(0.0030)	(0.0006)	(0.0026)	(0.0026)	(0.0008)
	0.30	0.2927	0.2934	0.2999	0.3022	0.3015	0.2985
		(0.0031)	(0.0031)	(0.0005)	(0.0033)	(0.0033)	(0.0008)
	0.45	0.4413	0.4424	0.4496	0.3048	0.3034	0.2990
		(0.0026)	(0.0026)	(0.0004)	(0.0041)	(0.0041)	(0.0008)
	0.60	0.5912	0.5927	0.6004	0.3072	0.3046	0.2983
		(0.0020)	(0.0019)	(0.0003)	(0.0058)	(0.0059)	(0.0009)
	0.75	0.7396	0.7416	0.7500	0.3149	0.3095	0.2980
		(0.0013)	(0.0013)	(0.0001)	(0.0090)	(0.0092)	(0.0009)

experiments were performed using the R programing language; see http://www.r-project.org. . . . . The CML estimates of α and λ were obtained by maximizing the conditional log-likelihood function using the BFGS quasi-Newton nonlinear optimization algorithm with numerical derivatives. As initial values for the algorithm, we considered the estimates obtained by CLS method. We computed 5000 replications and the sample sizes considered were $T = 50$, 100, 200 and 400. For the values of parameters, we considered $\alpha = 0.15, 0.3, 0.45, 0.6, 0.75$ and $\lambda = 0.3, 0.6$.

Tables 4 and 5 present the empirical means and mean squared errors of the estimates of the parameters of the BINAR(1) process. From the results we conclude that CML estimates present the smaller bias, as expected since it uses the whole information of the distribution, and the bias tends to zero for all estimators as the sample size

Table 5. Empirical means and mean squared errors (in parentheses) of the estimates of the parameters for $\lambda = 0.6$ and some values of α and T.

		Estimator of α			Estimator of λ		
T	α	$\hat{\alpha}_{YW}$	$\hat{\alpha}_{CLS}$	$\hat{\alpha}_{CML}$	$\hat{\lambda}_{YW}$	$\hat{\lambda}_{CLS}$	$\hat{\lambda}_{CML}$
50	0.15	0.1186	0.1210	0.1553	0.5926	0.5902	0.5826
		(0.0173)	(0.0182)	(0.0031)	(0.0095)	(0.0103)	(0.0060)
	0.30	0.2588	0.2645	0.3049	0.5975	0.5930	0.5826
		(0.0186)	(0.0190)	(0.0031)	(0.0113)	(0.0124)	(0.0061)
	0.45	0.3971	0.4059	0.4514	0.6075	0.6001	0.5845
		(0.0185)	(0.0180)	(0.0024)	(0.0119)	(0.0135)	(0.0062)
	0.60	0.5328	0.5449	0.6006	0.6239	0.6114	0.5832
		(0.0177)	(0.0162)	(0.0015)	(0.0149)	(0.0652)	(0.0063)
	0.75	0.6701	0.6864	0.7496	0.6551	0.6289	0.5838
		(0.0171)	(0.0145)	(0.0007)	(0.0202)	(0.0335)	(0.0064)
100	0.15	0.1323	0.1334	0.1525	0.5978	0.5971	0.5929
		(0.0095)	(0.0096)	(0.0015)	(0.0048)	(0.0049)	(0.0027)
	0.30	0.2774	0.2803	0.3008	0.5996	0.5977	0.5928
		(0.0097)	(0.0097)	(0.0014)	(0.0058)	(0.0060)	(0.0029)
	0.45	0.4219	0.4263	0.4507	0.6031	0.5994	0.5916
		(0.0092)	(0.0089)	(0.0012)	(0.0071)	(0.0075)	(0.0030)
	0.60	0.5645	0.5711	0.5998	0.6116	0.6053	0.5909
		(0.0081)	(0.0075)	(0.0008)	(0.0083)	(0.0088)	(0.0031)
	0.75	0.7085	0.7165	0.7496	0.6314	0.6197	0.5930
		(0.0066)	(0.0059)	(0.0003)	(0.0112)	(0.0121)	(0.0030)
200	0.15	0.1417	0.1424	0.1514	0.5986	0.5982	0.5965
		(0.0052)	(0.0052)	(0.0007)	(0.0024)	(0.0024)	(0.0013)
	0.30	0.2889	0.2903	0.3017	0.5997	0.5988	0.5959
		(0.0051)	(0.0051)	(0.0007)	(0.0029)	(0.0029)	(0.0014)
	0.45	0.4340	0.4364	0.4503	0.6025	0.6007	0.5956
		(0.0046)	(0.0045)	(0.0006)	(0.0034)	(0.0035)	(0.0014)
	0.60	0.5826	0.5855	0.5997	0.6051	0.6022	0.5956
		(0.0038)	(0.0036)	(0.0004)	(0.004)	(0.0045)	(0.0014)
	0.75	0.7303	0.7343	0.7500	0.6138	0.6079	0.5964
		(0.0028)	(0.0026)	(0.0002)	(0.0065)	(0.0067)	(0.0014)
400	0.15	0.1460	0.1463	0.1505	0.5993	0.5991	0.5984
		(0.0025)	(0.0025)	(0.0003)	(0.0012)	(0.0012)	(0.0007)
	0.30	0.2952	0.2959	0.3006	0.5993	0.5989	0.5979
		(0.0025)	(0.0025)	(0.0004)	(0.0014)	(0.0014)	(0.0007)
	0.45	0.4430	0.4441	0.4501	0.6008	0.6000	0.5985
		(0.0024)	(0.0024)	(0.0003)	(0.0019)	(0.0019)	(0.0007)
	0.60	0.5910	0.5925	0.6000	0.6039	0.6025	0.5989
		(0.0019)	(0.0018)	(0.0002)	(0.0023)	(0.0023)	(0.0007)
	0.75	0.7396	0.7414	0.7500	0.6074	0.6045	0.5981
		(0.0014)	(0.0013)	(0.0001)	(0.0036)	(0.0036)	(0.0007)

increases, since that the estimators are unbiased asymptotically. Although the CML has the best result, the CLS estimator is more indicated because it has closed form and its performance is very close to that of CML estimator. Furthermore, as suggested by the referees, Tables 6 and 7 present the computation time of the estimates of the parameters of the BINAR(1) process. As can be seen from results, the CLS estimator has a little computational cost in contrast with the CML estimator.

6. Empirical illustrations

Here, we present two applications of the Borel INAR(1) model to a real data set. First, in Section 6.1, we consider an underdispersed real data set and second, in Subsection

Table 6. Execution time of the estimates of the parameters for $\lambda = 0.3$ and some values of α and T.

T	α	$\hat{\alpha}_{YW}$	$\hat{\alpha}_{CLS}$	$\hat{\alpha}_{CML}$
50	0.15	4.5517 secs	3.2687 secs	1.8598 mins
	0.30	5.0213 secs	3.1161 secs	1.2292 mins
	0.45	5.2654 secs	3.1735 secs	1.1202 mins
	0.60	4.3496 secs	3.1775 secs	1.2647 mins
	0.75	5.4105 secs	3.2733 secs	1.8191 mins
100	0.15	5.3823 secs	3.9171 secs	1.7825 mins
	0.30	5.1073 secs	3.8373 secs	2.1895 mins
	0.45	6.0549 secs	3.8525 secs	2.8829 mins
	0.60	5.3331 secs	3.7598 secs	3.8640 mins
	0.75	6.1555 secs	3.7512 secs	4.8715 mins
200	0.15	7.9261 secs	5.7598 secs	6.3217 mins
	0.30	6.9359 secs	5.3239 secs	7.8635 mins
	0.45	6.6425 secs	5.8507 secs	5.1088 mins
	0.60	6.2880 secs	5.3869 secs	5.6261 mins
	0.75	6.7210 secs	5.3205 secs	7.4023 mins
400	0.15	8.4952 secs	8.4098 secs	4.9553 mins
	0.30	8.9713 secs	8.4360 secs	8.6206 mins
	0.45	9.3678 secs	8.3646 secs	11.432 mins
	0.60	9.1145 secs	8.3926 secs	10.761 mins
	0.75	10.653 secs	8.3949 secs	22.149 mins

Table 7. Execution time of the estimates of the parameters for $\lambda = 0.6$ and some values of α and T.

T	α	$\hat{\alpha}_{YW}$	$\hat{\alpha}_{CLS}$	$\hat{\alpha}_{CML}$
50	0.15	6.6341 secs	6.7016 secs	3.0916 mins
	0.30	6.3741 secs	5.2456 secs	1.2480 mins
	0.45	7.7133 secs	5.2754 secs	1.6473 mins
	0.60	9.6796 secs	7.4349 secs	2.0438 mins
	0.75	9.7923 secs	8.5439 secs	4.7131 mins
100	0.15	8.0072 secs	8.8296 secs	2.5656 mins
	0.30	11.347 secs	6.5589 secs	3.9113 mins
	0.45	12.294 secs	6.7440 secs	3.8778 mins
	0.60	9.7499 secs	9.5958 secs	4.0087 mins
	0.75	13.119 secs	13.698 secs	6.9888 mins
200	0.15	15.205 secs	12.831 secs	9.1867 mins
	0.30	11.054 secs	8.0066 secs	4.9965 mins
	0.45	16.474 secs	7.0033 secs	4.8192 mins
	0.60	9.5122 secs	6.8397 secs	7.1441 mins
	0.75	15.055 secs	6.9684 secs	21.530 mins
400	0.15	21.3451 secs	10.687 secs	9.9275 mins
	0.30	19.5723 secs	10.896 secs	15.310 mins
	0.45	19.2202 secs	10.731 secs	20.701 mins
	0.60	16.3232 secs	11.052 secs	16.913 mins
	0.75	19.4273 secs	11.124 secs	26.105 mins

6.2, we fitted an overdispersed real data set to BINAR(1) model. We considered the CML method to estimate the unknown parameters of the models in both cases. After, we considered the $\hat{\lambda}_{CLS}$ estimator and the asymptotic distribution given in Theorem 4.1. So, we made a hypothesis test in both applications to test the hypothesis $H_0 : \lambda = 3/2 - \sqrt{5}/2$ against $H_1 : \lambda < 3/2 - \sqrt{5}/2$ (underdispersed model) and $H_0 : \lambda = 3/2 - \sqrt{5}/2$ against $H_1 : \lambda > 3/2 - \sqrt{5}/2$ (overdispersed model). The required numerical evaluations were carried out using the R programing language as in Section 5.

Table 8. Counts of vagrancy data descriptive statistics.

Minimum	Median	Mean	Variance	Skewness	Kurtosis	$\hat{\rho}(1)$	Maximum
1	1	1.3402	0.4358	2.2851	8.5345	0.1959	4

6.1. Modeling underdispersion

Now, we present an application of the Borel INAR(1) model to the real data set in the underdispersed case. The data set is given by Bakouch and Ristić (2010) as an application of the ZTPINAR(1) process. Their original data are counts of vagrancy in the 64th police car beat in Pittsburgh, during one month. The data consists of 144 observations, starting in January 1990 and ending in December 2001. It can be obtained from the crime data section of the forecasting principles site (http://www.forecastingprinciples.com. . . . .). With the aim to use the ZTPINAR(1) model, that authors transformed the series adding 1 to each observation. Here we consider the transformed series too.

In Table 8, we present some empirical descriptive measures for the data set of counts of vagrancy, which include central tendency statistics, variance, skewness and kurtosis, among others. We can observe that the empirical mean is much greater than the empirical variance of data set, showing the data is empirically under dispersed.

The time series data and their sample autocorrelation and partial autocorrelation are displayed in Figure 3. From Figure 3, we can see that a first order autoregressive model may be appropriate for the given data series, because of the clear cutoff after lag 1 in the partial autocorrelations. Furthermore, the behavior of the series indicates that it may be mean stationary. We compare our model with the shifted models: Truncated Poisson INAR(1) model (Bourguignon and Vasconcellos 2015) and ZTPINAR(1) model (Bakouch and Ristić 2010). Table 9 presents the CML estimates with corresponding standard errors in parentheses, Akaike information criterion (AIC) and the root mean square of differences between observations and predicted values (RMS). From Table 9, we observe that the BINAR(1), truncated Poisson INAR(1) and ZTPINAR(1) models are competitive. The BINAR(1) model showed better goodness-of-fit statistics compared with the truncated Poisson INAR(1) model. From the results, we conclude that the BINAR(1) is adequate for this time series.

The sample autocorrelations of the residuals from Borel INAR(1) process are displayed in Figure 4.

Now, our interest is to test the null hypothesis $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda < \lambda_0$, where $\lambda_0 = (3-\sqrt{5})/2 = 0.382$ on significance level $\delta = 0.05$. In other words, we want to test the null hypothesis of an equidispersed model against an underdispersed model. For this, we considered the CLS estimator and the asymptotic distribution given in Theorem 4.1. In Table 10 we have the empirical results of the CLS estimators to λ and α, the standard deviation of the CLS estimator of λ, the test statistic, the p-value and the critical value.

With the results given in Table 10, we have that $\hat{\lambda}_{\text{CLS}} = 0.0733 < 0.5082 = \lambda_0 + q_\delta\sqrt{\hat{\sigma}^2_{\lambda_0}/144}$, therefore we reject the null hypothesis $H_0 : \lambda = \lambda_0$ in favor of the alternative hypothesis $H_1 : \lambda < \lambda_0$ with a significance level $\delta = 0.05$. So, we have evidence for an underdispersed model, as expected. Alternatively, from a p-value analysis we have that p-value $= 0.0006 < 0.05 = \delta$, confirming the result of the hypothesis test.

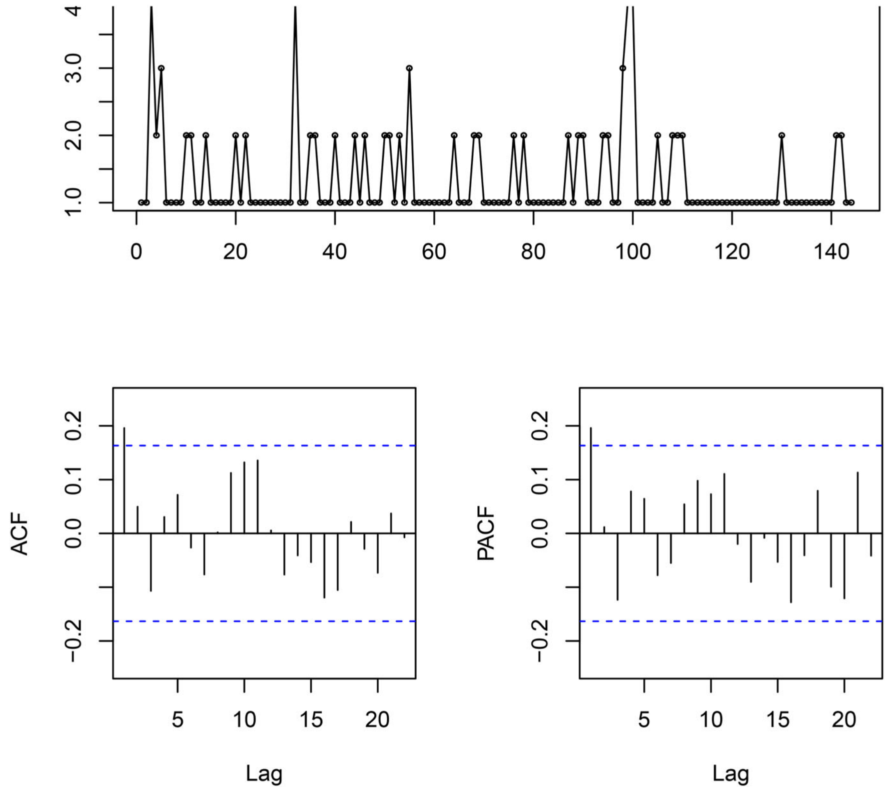


Figure 3. The sample path of the time series counts of vagrancy, the autocorrelation function and the partial autocorrelation function.

Table 9. Estimated parameters (with corresponding standard errors in parentheses), RMS and AIC.

Model	CML estimates	RMS	AIC
BINAR(1)	$\hat{\alpha} = 0.1156$ (0.0468) $\hat{\lambda} = 0.1577$ (0.0493)	0.6489	218.9168
Truncated Poisson INAR(1)	$\hat{\alpha} = 0.0867$ (0.0600) $\hat{\lambda} = 0.4225$ (0.1514)	0.6507	221.2132
ZTPINAR(1)	$\hat{\alpha} = 0.3886$ (0.1394) $\hat{\lambda} = 0.6216$ (0.1025)	0.6468	217.8503

6.2. Modeling overdispersion

The data set of the second application consists of 144 observations are from MacDonald and Zucchini (1997). Consider the series of weekly sales (in integer units) of a particular soap product in a supermarket. The data are taken from a database provided by the Kilts Center for Marketing, Graduate School of Business of the University of Chicago, at:http://gbswww.uchicago.edu/kilts/research/db/dominicks. (The product is 'Zest White Water 15 oz.', with code 3700031165). In order to use the BINAR model, as made by Bakouch and Ristić (2010), we transformed the series adding 1 to each observation.

In Table 11, we present some descriptive measures for the data set of weekly sales, which include central tendency statistics, variance, skewness and kurtosis, among others.

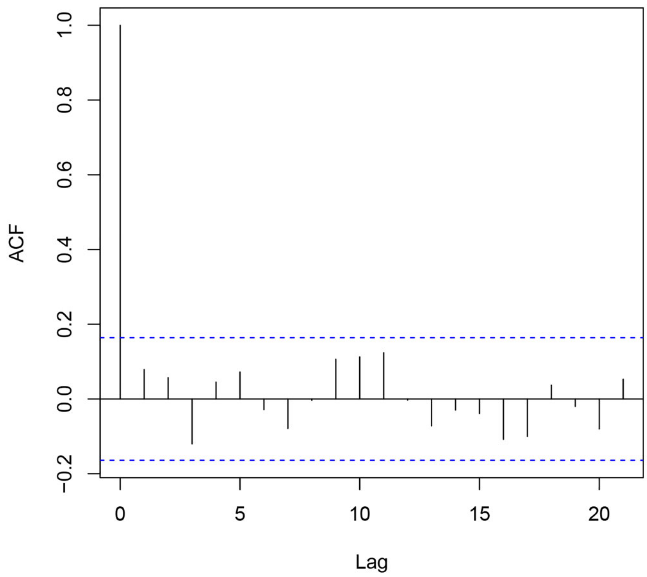


Figure 4. Sample autocorrelations of the residuals obtained from Borel INAR(1) model.

Table 10. Empirical results from hypothesis test apply to time series counts of vagrancy.

Data set size	$\hat{\lambda}_{\text{CLS}}$	$\hat{\alpha}_{\text{CLS}}$	s.d.$_{\hat{\lambda}_{\text{CLS}}}$	test statistic	p-value	critical value
144	0.0733	0.1962	1.1469	−3.2290 1	0.0006	0.5082

We can observe that the mean is smaller than the variance of the data set, showing overdispersion.

In Figure 5, the time series data and their sample autocorrelation and partial autocorrelation are displayed. Figure 5 shows that a first order autoregressive model may be appropriate for the given data series. Furthermore, the behavior of the series indicates that it may be mean stationary. The sample autocorrelations of the residuals from Borel INAR(1) process are displayed in Figure 6.

We compare our model with the Truncated Poisson INAR(1) model (Bourguignon and Vasconcellos 2015) and the ZTPINAR(1) model (Bakouch and Ristić 2010). Table 12 presents the CML estimates with corresponding standard errors in parentheses, AIC and the RMS. From Table 12, we observe that the BINAR(1) model showed better goodness-of-fit statistics compared with the other models. From the results, we conclude that the BINAR(1) is adequate for this time series.

As previously in underdispersed case, now our interest is to test the null hypothesis $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda > \lambda_0$, where $\lambda_0 = \left(3 - \sqrt{5}\right)/2 = 0.382$ on significance level $\delta = 0.05$, that is, we want test the null hypothesis of an equidispersed model against an overdispersed model. Again, we considered the CLS estimator to λ and the asymptotic distribution given in Theorem 4.1. In Table 13, we have the empirical results of the CLS estimator to λ and α, the standard deviation of CLS estimator, the test statistic, p-value and the critical value.

Table 11. Series of weekly sales descriptive statistics.

Minimum	Median	Mean	Variance	Skewness	Kurtosis	$\hat{\rho}(1)$	Maximum
1	5	6.4421	15.4012	1.4998	5.7926	0.3923	23

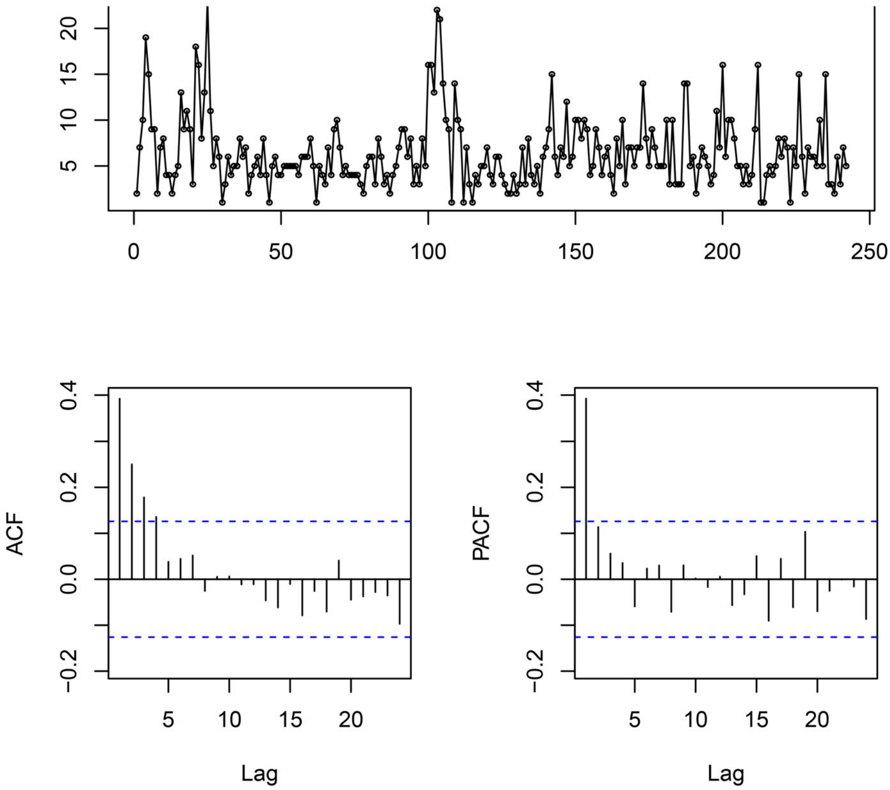


Figure 5. The sample path of the time series of weekly sales, the autocorrelation function and the partial autocorrelation function.

With the results given in Table 13, we have that $\hat{\lambda}_{\text{CLS}} = 0.7455 > 0.5118 = \lambda_0 + q_\delta\sqrt{\hat{\sigma}^2_{\lambda_0}/144}$, therefore we reject the null hypothesis $H_0 : \lambda = \lambda_0$ in favor of the alternative hypothesis $H_1 : \lambda < \lambda_0$ with a significance level $\delta = 0.05$. So, we have evidence for an overdispersed model, as expected. Alternatively, from a p-value analysis, we have a extremely small p-value $= 4.5519 \times 10^{-15} < 0.05 = \delta$, confirming the result of the hypothesis test.

7. Concluding remarks

In this paper we introduced the shifted first order integer-valued autoregressive process with Borel innovations. This model was based on the binomial thinning with Borel distribution innovations. The Borel INAR(1) model is suited to equidispersed, underdispersed and overdispersed data counts set. The basic properties of this model are obtained. The closed expressions from CLS and YW estimators of unknown parameters are derived and we obtain of their asymptotic distribution. An hypothesis test based on

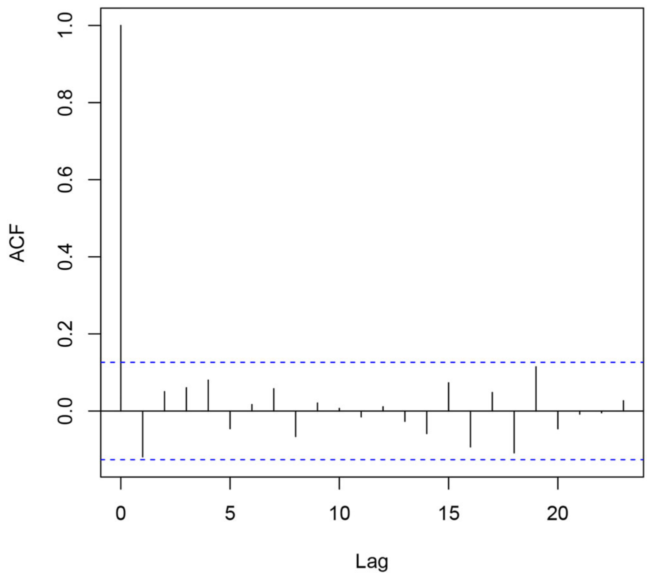


Figure 6. Sample autocorrelation of the residuals obtained from Borel INAR(1) model.

Table 12. Estimated parameters (with corresponding standard errors in parentheses), RMS and AIC.

Model	CML estimates	RMS	AIC
BINAR(1)	$\hat{\alpha} = 0.4647$ (0.0211) $\hat{\lambda} = 0.7112$ (0.0305)	3.6093	1317.036
Truncated Poisson INAR(1)	$\hat{\alpha} = 0.2222$ (0.0335) $\hat{\lambda} = 4.9934$ (0.2564)	3.6597	1328.947
ZTPINAR(1)	$\hat{\alpha} = 0.2433$ (0.0361) $\hat{\lambda} = 6.4567$ (0.2109)	3.6457	1324.429

Table 13. Empirical results from hypothesis test applied to time series of weekly sales.

Data set size	$\hat{\lambda}_{CLS}$	$\hat{\alpha}_{CLS}$	s.d.$_{\hat{\lambda}_{CLS}}$	test statistic	*p*-value	critical value
242	0.7455	0.3925	0.7295	7.7513	4.5519×10^{-15}	0.5118

the asymptotic distribution of CLS estimators was formulated to test equidispersed against overdispersed or underdispersed model. We made a test power analysis by simulating the rejected rate for different sample sizes considering the true and estimated value of α and the asymptotic distribution. Two applications were presented showing that the Borel INAR(1) model is suited to shifted underdispersed and overdispersed data sets and the results of a hypothesis test confirmed the assessment.

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Appendix

Proof of Theorem 4.1. The Borel INAR(1) process $\{X_t\}_{t\in\mathbb{Z}}$ is a branching process with immigration whose offspring distribution is Bernoulli and the immigration process has Borel distribution, both having all moments finite. By Klimko and Nelson (1978), the CLS estimators of α and μ_ϵ are strongly consistent and their asymptotic joint distribution can be obtained applying the results from Section 5 in the same paper. We have,

$$\sqrt{T}(\hat{\alpha}_{\text{CLS}} - \alpha, \hat{\mu}_{\epsilon;\text{CLS}} - \mu_\epsilon) \xrightarrow{d} N(0, \mathbf{\Sigma}^*)$$

where $\hat{\alpha}_{\text{CLS}}$ is give in (8) and $\hat{\mu}_{\epsilon;\text{CLS}}$ is given by the expression

$$\hat{\mu}_{\epsilon;\text{CLS}} = \frac{1}{T}\left(\sum_{t=1}^{T} X_t - \hat{\alpha}_{\text{CLS}} \sum_{t=2}^{T} X_{t-1}\right)$$

with $\mathbf{\Sigma}^* = (\sigma^*_{ij})$ being obtained from the results given in Klimko and Nelson (1978), Section 5, by

$$\sigma^*_{11} = (1-\alpha^2) + \frac{\alpha(1-\alpha)(1-\alpha^2)^2(1-\lambda)^5}{(1-\alpha^3)\left[\lambda + \alpha(1-\lambda)^2\right]^2}\left[\frac{\alpha}{1-\alpha} + \frac{3\alpha^3}{1+\alpha} + \frac{3\alpha^2\lambda}{(1+\alpha)(1-\lambda)^2} + \frac{1+2\lambda}{(1-\lambda)^4}\right]$$

$$\begin{aligned}
\sigma^*_{12} &= \alpha(1-\alpha) - \frac{\lambda(1-\alpha)}{(1-\lambda)\left[\lambda + \alpha(1-\lambda)^2\right]} - \frac{\alpha(1+\alpha)(1-\lambda)}{\lambda + \alpha(1-\lambda)^2} \\
&\quad - \left[(1+\alpha) + 3\alpha^2(1-\alpha)\right]\frac{\alpha^2(1-\alpha^2)(1-\lambda)^5}{\left[\lambda + \alpha(1-\lambda)^2\right]^2} \\
&\quad - \left[3\alpha^2\lambda(1-\lambda)^2 + (1+2\lambda)(1+\alpha)\right]\frac{\alpha(1-\alpha)^2(1+\alpha)(1-\lambda)}{\left[\lambda + \alpha(1-\lambda)^2\right]^2}
\end{aligned}$$

$$\begin{aligned}
\sigma^*_{22} &= \frac{1}{(1-\lambda)^2(1-\alpha)^2} - \frac{\alpha}{1-\lambda} + \frac{\lambda}{(1-\lambda)^3} + \frac{\alpha(1+\alpha)}{(1-\alpha)\left[\lambda + \alpha(1-\lambda)^2\right]} \\
&\quad + \frac{\alpha(1-\alpha)(1+\alpha)^2(1-\lambda)^3}{(1-\alpha^3)\left[\lambda + \alpha(1-\lambda)^2\right]^2}\left[\frac{\alpha}{1-\alpha} + \frac{3\alpha^3}{1+\alpha} + \frac{3\lambda\alpha^2}{(1+\alpha)(1-\lambda)^2} + \frac{1+2\lambda}{(1-\lambda)^4}\right]
\end{aligned}$$

Now, we define the function $f := \mathbb{R}^2_* \to \mathbb{R}^2$, as follows

$$f = (f_1, f_2)^\top = (f_1(x_1, x_2), f_2(x_1, x_2)) := \left(x_1, 1 - \frac{1}{x_2}\right)$$

where $\mathbb{R}^2_* := \{(x_1, x_2) \in \mathbb{R}^2; x_2 \neq 0\}$. We have that the f function is continuous and differentiable at its domain. According to the delta method, considering $\boldsymbol{\theta} = (\alpha, \mu_\epsilon)$ and $\mathbf{Y_T} = (\hat{\alpha}_{\text{CLS}}, \hat{\mu}_{\epsilon;\text{CLS}})$, we have

$$\sqrt{T}(f(\mathbf{Y_T}) - f(\theta)) \xrightarrow{d} N(0, \mathrm{D}\mathbf{\Sigma}^*\mathrm{D}^\top)$$

where $\mathbf{D}$ is the Jacobian matrix of f with respect to $\boldsymbol{\theta}$.

The first-order partial derivatives of f is given by

$$\frac{\partial f_1(x_1, x_2)}{\partial x_1} = 1, \quad \frac{\partial f_1(x_1, x_2)}{\partial x_2} = 0, \quad \frac{\partial f_2(x_1, x_2)}{\partial x_1} = 0, \quad \frac{\partial f_2(x_1, x_2)}{\partial x_2} = \frac{1}{y^2}$$

thus, the Jacobian matrix of f with respect to $\boldsymbol{\theta}$ is

$$\mathbf{D} = \begin{pmatrix} 1 & 0 \\ 0 & \dfrac{1}{\mu_\epsilon^2} \end{pmatrix}$$

We have,

$$\mathbf{\Sigma} = \mathbf{D}\Sigma^*\mathbf{D}^\top = \begin{pmatrix} \sigma^*_{11} & \dfrac{1}{\mu_\epsilon^2}\sigma^*_{12} \\ \dfrac{1}{\mu_\epsilon^2}\sigma^*_{12} & \dfrac{1}{\mu_\epsilon^4 \sigma^*_{22}} \end{pmatrix}$$

Finally, we obtain

$$\sigma_{11} = (1-\alpha^2) + \frac{\alpha(1-\alpha)(1-\alpha^2)^2(1-\lambda)^5}{(1-\alpha^3)\left[\lambda+\alpha(1-\lambda)^2\right]^2}\left[\frac{\alpha}{1-\alpha}+\frac{3\alpha^3}{1+\alpha}+\frac{3\alpha^2\lambda}{(1+\alpha)(1-\lambda)^2}+\frac{1+2\lambda}{(1-\lambda)^4}\right]$$

$$\begin{aligned}\sigma_{12} &= \alpha(1-\alpha)(1-\lambda)^2 - \frac{\lambda(1-\alpha)(1-\lambda)}{\lambda+\alpha(1-\lambda)^2} - \frac{\alpha(1+\alpha)(1-\lambda)^3}{\lambda+\alpha(1-\lambda)^2} \\ &\quad - \left[(1+\alpha)+3\alpha^2(1-\alpha)\right]\frac{\alpha^2(1-\alpha^2)(1-\lambda)^7}{\left[\lambda+\alpha(1-\lambda)^2\right]^2} \\ &\quad - \left[3\alpha^2\lambda(1-\lambda)^2+(1+2\lambda)(1+\alpha)\right]\frac{\alpha(1-\alpha)^2(1+\alpha)(1-\lambda)^3}{\left[\lambda+\alpha(1-\lambda)^2\right]^2}\end{aligned}$$

$$\begin{aligned}\sigma_{22} &= \left\{1-\alpha\left[\alpha\lambda+(1-2\alpha)(1+\lambda^2)+\alpha^2(1-\lambda)^2\right]\right\}\frac{1-\lambda}{(1-\alpha)^2} \\ &\quad + \frac{\alpha(1+\alpha)(1-\lambda)^4}{(1-\alpha)(\lambda+\alpha(1-\lambda)^2)} + \frac{3\alpha^3(1-\lambda)^5(1-\alpha^2)}{(1-\alpha^3)(\lambda+\alpha(1-\lambda)^2)} \\ &\quad + \left[(1-\alpha)(1+2\lambda)+\alpha(1-\lambda)^4\right]\frac{\alpha(1+\alpha)^2(1-\lambda)^3}{(1-\alpha^3)\left[\lambda+\alpha(1-\lambda)^2\right]^2}\end{aligned}$$

Proof of Theorem 4.2. The proof is done in the same way of Freeland and McCabe (2005). The proof to show that $\sqrt{T}(\hat{\alpha}_{\text{CLS}}-\hat{\alpha}_{\text{YW}})$ is $o_p(1)$ in Borel INAR(1) process is identically the proof in Poisson INAR(1) process. For α estimators, we have:

Let,

$$D_{\text{CLS}} = \frac{1}{T}\left[\sum_{t=2}^{T}X_{t-1}^2 - \frac{1}{T-1}\left(\sum_{t=2}^{T}X_{t-1}\right)^2\right] \quad \text{and} \quad D_{\text{YW}} = \frac{1}{T}\sum_{t=1}^{T}(X_t-\bar{X})^2 = \frac{1}{T}\sum_{t=1}^{T}X_t^2 - \bar{X}^2$$

thus,

$$\begin{aligned}\sqrt{T}(\hat{\alpha}_{\text{YW}}-\hat{\alpha}_{\text{CLS}}) &= \sqrt{T}\left[\frac{D_{CLS}\left(\frac{1}{T}\sum_{t=2}^{T}X_tX_{t-1} - \bar{X}\frac{1}{T}\sum_{t=2}^{T}X_t - \bar{X}\frac{1}{T}\sum_{t=2}^{T}X_{t-1} + \left(1-\frac{1}{T}\right)\bar{X}^2\right)}{D_{\text{CLS}}D_{\text{YW}}}\right] \\ &\quad -\sqrt{T}\left[\frac{D_{\text{YW}}\left(\frac{1}{T}\sum_{t=2}^{T}X_tX_{t-1} - \frac{1}{T(T-1)}\sum_{t=2}^{T}X_t\sum_{t=2}^{T}X_{t-1}\right)}{D_{\text{CLS}}D_{\text{YW}}}\right],\end{aligned}$$

After some algebra, we conclude

$$\begin{aligned}\sqrt{T}(\hat{\alpha}_{\text{YW}}-\hat{\alpha}_{\text{CLS}}) &= \frac{(D_{\text{CLS}}-D_{\text{YW}})}{D_{\text{CLS}}D_{\text{YW}}}\cdot\frac{\sum_{t=2}^{T}X_tX_{t-1}}{\sqrt{T}} - \frac{\left(D_{\text{CLS}}\sum_{t=2}^{T}\frac{X_t}{T} - D_{\text{YW}}\sum_{t=2}^{T}\frac{X_t}{T-1}\right)}{D_{\text{CLS}}D_{\text{YW}}}\cdot\frac{\sum_{t=1}^{T}X_t}{\sqrt{T}} \\ &\quad - \frac{D_{\text{YW}}}{D_{\text{CLS}}D_{\text{YW}}}\cdot\sum_{t=2}^{T}\frac{X_t}{T-1}\frac{X_T}{\sqrt{T}} + \frac{D_{\text{CLS}}}{D_{\text{CLS}}D_{\text{YW}}}\cdot\bar{X}\frac{(X_T-\bar{X})}{\sqrt{T}} \\ &= o_p(1)O_p(1) - o_p(1)O_p(1) - O_p(1)O_p(1)o_p(1) + O_p(1)O_p(1)o_p(1) = o_p(1)\end{aligned}$$

For λ estimators, define

$$D_{\mathrm{CLS}} = \frac{1}{T}\sum_{t=2}^{T} X_t - \hat{\alpha}_{\mathrm{CLS}}\frac{1}{T}\sum_{t=2}^{T} X_{t-1} \quad \text{and} \quad D_{\mathrm{YW}} = \bar{X}(1-\hat{\alpha}_{\mathrm{YW}}).$$

Note that D_{CLS} and D_{YW} converges to the same non zero constant $\mu(1-\alpha)$.

We have,

$$\begin{aligned}
\sqrt{T}(\hat{\lambda}_{\mathrm{YW}} - \hat{\lambda}_{\mathrm{CLS}}) &= \sqrt{T}\left[\frac{1}{\bar{X}(1-\hat{\alpha}_{\mathrm{YW}})} - \frac{T-1}{\sum_{t=2}^{T} X_t - \hat{\alpha}_{\mathrm{CLS}}\sum_{t=2}^{T} X_{t-1}}\right] \\
&= \sqrt{T}\left\{\frac{\frac{1}{T}\sum_{t=2}^{T} X_t - \hat{\alpha}_{\mathrm{CLS}}\frac{1}{T}\sum_{t=2}^{T} X_{t-1} - \left(1-\frac{1}{T}\right)\left[\bar{X}(1-\hat{\alpha}_{\mathrm{YW}})\right]}{D_{\mathrm{CLS}} D_{\mathrm{YW}}}\right\}.
\end{aligned}$$

After some algebra, we have

$$\begin{aligned}
\sqrt{T}(\hat{\lambda}_{\mathrm{YW}} - \hat{\lambda}_{\mathrm{CLS}}) &= \frac{\frac{1}{\sqrt{T}}(X_T\hat{\alpha}_{\mathrm{CLS}} - X_1) - \sqrt{T}(\hat{\alpha}_{\mathrm{CLS}} - \hat{\alpha}_{\mathrm{YW}})\bar{X} + \frac{\bar{X}}{\sqrt{T}}(1-\hat{\alpha}_{\mathrm{YW}})}{D_{\mathrm{CLS}} D_{\mathrm{YW}}} \\
&= o_p(1) - o_p(1) + o_p(1) = o_p(1).
\end{aligned}$$